



OIL INDUSTRY SAFETY DIRECTORATE

**नॉन-पिगेबल तटीय हाइड्रोकार्बन पाइपलाइन
ओआईएसडी – मानक –233**

**NON PIGGABLE ONSHORE HYDROCARBON
PIPELINE
OISD-STD-233**

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FOR RESTRICTED CIRCULATION ONLY

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PREFACE

Indian petroleum industry is the energy lifeline of the nation and its continuous performance is essential for prosperity of the country. As the industry deals with inherently inflammable substances throughout its value chain – upstream, midstream and downstream – Safety is of paramount importance to this industry as safe performance at all times can ensure optimum return of these national assets and resources including sustainability.

To ensure proper implementation of the various aspects of safety in the oil & gas industry, the Government of India, vide Resolution No. R-13013/4/84-OR-1 dated 10th January, 1986 set up a Safety Council at the apex, under the administrative control of the Ministry of Petroleum & Natural Gas (MoP&NG). To assist the Safety Council, a technical directorate, namely the Oil Industry Safety Directorate (OISD) under the aegis of MoP&NG, has been entrusted with the responsibility of formulating standards, overseeing its implementation through safety audits to enhance safety level in the oil & gas industry.

OISD has developed a rigorous, multi-layer, iterative and participative approach for development of standards. OISD develops and revises Standards, Guidelines & Recommended Practices for the oil and gas sector through a participative process involving all the stakeholders, drawing inputs from national and international standards and adapting them by leveraging the experience of the functional committee. These standards cover inbuilt design safety, asset integrity and best operating practices in the field of production, processing, storage and transport of petroleum.

The participative process followed in standard formulation has resulted in excellent level of compliance by the industry. It also goes to prove the old adage that self-regulation is the best regulation. The quality and relevance of OISD standards had been further endorsed by their adoption/ reference in various statutory rules and regulations of the land.

OISD and industry together strive to achieve nil incident in the entire hydrocarbon value chain. We at OISD, are confident that the provisions of this standard, when implemented in totality, would go a long way in ensuring safe operation in the oil and gas industry.

**Executive Director
Oil Industry Safety
Directorate**

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Wherever Acts/ Rules/ Regulation and National/ International Standards are mentioned in the standard, the same relates to in-vogue version of such documents. Whenever there is any difference between the statutory norm and OISD standard, more stringent requirement should be followed. However, in case of any difference in the requirement between OISD standards, the requirement as specified in the standard released at a later stage shall prevail.

The industry has to carry out gap analysis and make necessary changes in their system in line with the revised standard and the requirement should be implemented within a time frame of 1 year (from the date of release of revised standard).

Though every effort has been made to assure the accuracy and reliability of the information contained in these documents, OISD hereby expressly disclaims any liability or responsibility for loss or damage resulting from the use of OISD standard.

The standards would be reviewed periodically based on the decision in the Steering Committee/ Safety Council, industry request, technological development, recommendations of high level committee and accident investigation reports. Generally, OISD standards are reviewed and revised, reaffirmed or withdrawn at least every ten years.

Suggestions for revision are invited from the oil and gas industry and should be submitted to OISD at **ps-ed.oisd@gov.in**.

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Mukesh Kumar	Chief General Manager	ONGC
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Siddharth Ojah	Chief Engineer	OIL (up to: 07.05.24)
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Jitesh Dixit	Addl. Director (E&P)	Oil Industry Safety Directorate
Devendra Kumar	Joint Director (LPG)	Oil Industry Safety Directorate
MEMBER COORDINATOR		
Abhishek Srivastava	Joint Director (Pipeline)	Oil Industry Safety Directorate

OIDSD also recognizes the support & cooperation of several other experts from industry, who contributed to the preparation, review and finalization of this document.

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Non Piggable Onshore Hydrocarbon Pipelines

1.0 INTRODUCTION

Safety in petroleum pipeline comes through continuous efforts at all stages and it can be ensured by observing that pipelines are designed, and constructed, as per recognized engineering standards and they are periodically inspected and maintained.

2.0 SCOPE

This standard is applicable for:

- a) Design, construction, operation, inspection and maintenance of onshore non piggable metallic pipelines of upstream hydrocarbon industry pipeline of sizes 2" NPS and above, laid between Wellhead and installation for transportation of well fluid, water/effluent disposal, gas injection, lift gas, water and polymer injection etc.
- b) Inspection and Integrity assessment of onshore metallic cross-country pipelines designed as per OISD-STD-141, OISD-STD-214 & OISD-STD-226 which are either non piggable or have become unpiggable.

This standard does not include:

- i) In plant process piping.
- ii) Jetty pipelines.
- iii) Cross country pipelines and city gas distribution pipelines (covered under respective OISD standard i.e. OISD-STD-141, OISD-STD-214 & OISD-STD-226)

This standard shall be applicable from the date of issuance mentioned on title page for all new projects / extension or expansion of existing system.

This standard shall be applicable from the date of issuance mentioned on title page for all new projects / extension or expansion of existing system. For new pipeline under construction, commissioning phase and also for existing pipelines with associated facilities under operation phase, the requirements related to Operation, Inspection & Maintenance, Repair, Abandonment, Preservation, Inspection and Integrity Assessment shall be applicable.

3.0 DEFINITION:

3.1 Non Piggable Pipeline:

A pipeline that does not have required facility to pig or a pipeline that cannot be pigged in their existing operating conditions and same has been approved by management of operating company.

3.2 Flow line/ Well fluid line:

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Pipeline that transports fluid between wells & associated facilities.

3.3 Feeder Line/ Collector line:

Any pipeline transporting processed fluids between various installation/ process facilities of the same operator in upstream sector.

3.4 Multiphase Fluids:

Multiphase fluids mean oil, gas or water in any combination produced from one or more oil wells or recombined oil well fluids that may have been separated in passing through treatment/processing facilities. For the purpose of this standard, multiphase fluids are considered to be low vapour pressure fluids.

3.5 Purging:

It is the process of replacing the atmospheric air within a container (pipeline, vessels, filters etc) by an inert substance in such a manner so as to prevent the formation of explosive mixture.

3.6 Shall

Shall indicates mandatory requirement.

3.7 Should

Should indicates that the provision is recommended as a good practice.

4.0 PIPELINE DESIGN

Design shall ensure public safety as a priority, considering conditions during construction, testing, commissioning and operating periods. It must be designed based on environmental and well fluid properties. Pipe diameter NPS 2" and above shall be used for flow lines. For all new carbon steel pipe nominal wall thickness including corrosion allowance shall not be less than 7 mm. Reference may be taken of section 5.1.6 of OISD-STD-141.

Design life of flow line should be based on Well's Production life cycle. Design shall consider all internal and external factors affecting the pipeline system.

The nominal wall thickness "tn" for the steel pipe shall be calculated as below:

$$t_n \geq t + \text{allowances}$$

t- Internal design pressure wall thickness of pipe.

i.e

$$t = \frac{P_i D}{2S}$$

Where,

D = Outside diameter of pipe, in.

P_i = Internal design gauge pressure, psi

S = Applicable allowable stress value, psi

= F x E x SMYS, psi

F = **Design Factor (As per Table-1, clause no. 5.1.6.2 of OISD-STD-141)**

E = Longitudinal joint factor, which for electric welded (EW), longitudinal seam submerged arc welded (LSAW), helical seam submerged arc welded (HSAW) and

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seamless types of pipes, manufactured in accordance with API specification 5L and considered as 1;

SMYS = Specified Minimum Yield Strength, psi.

4.1 MATERIAL

Carbon Steel line pipe for use in hydrocarbon pipeline system shall be Seamless, Electric Welded (HFW) or Longitudinal / Helical Seam Submerged Arc Welded (SAWL/SAWH) type conforming to Line pipe Specification API 5L Product Specification Level (PSL) - 2 or equivalent.

However, based on reservoir & flow/ fluid properties, operator may consider other suitable materials as per fluid properties with proper engineering documentation during design phase as per applicable international/ national standards.

5.0 EXTERNAL COATING FOR BURIED PIPELINE

Pipeline shall be externally coated with suitable anti corrosion coating e.g. 3-layer Polyethylene / Polypropylene or Fusion Bonded Epoxy coating or any other equivalent type of coating. Anticorrosion coating shall be selected considering the varying ground conditions found during soil resistivity and soil Microbial, Ionic loading survey carried out along the pipeline route. Selected coating shall meet the following long term functional requirement:

- Provide electrical isolation between the external surface of the pipeline and environment.
- Have sufficient adhesion to pipe surface to effectively resist under film ingress/ migration of moisture.
- Have sufficient ductility to prevent cracking during field bending and to have a suitable repair methodology for field repair, if required after bending.
- Have adequate mechanical properties to resist damage during handling, laying and operational stresses and have a suitable repair methodology for field repair.
- Suitable for the pipeline operating temperature range.

5.1 COATING SPECIFICATION:

Specification for external anticorrosion coating shall include as a minimum, the following requirements:

- Type of the coating system.
- Minimum coating thickness (in case of multilayer coating, thickness of each layer should be specified).
- Properties of base materials.
- Properties of as applied coating.
- Operating temperature range which the coating must withstand without deterioration of its physical and electrical properties.
- Minimum surface preparation and coating application requirements.
- Adhesion requirements for materials, application and curing, including possible requirements for health, safety and environmental aspects
- Inspection and Testing requirement (for holiday detection & repair)
- Repair of damaged coating at site for field repair.

6.0 CONSTRUCTION

6.1 General:

- a) Construction, repair and replacement of flowlines shall be in accordance with the requirement of this Section. It is not intended herein that all construction items be covered in full detail, whether covered specifically or not, all construction and material shall be in accordance with good engineering, safety and proven pipeline practice.
- b) Minimum cover for Flowlines shall be as per Table-5 of clause No. 11.5 of OISD-STD-141, unless construction above ground is found to be necessary for exceptional technical reasons or constraints. Operating company must have a well-defined construction methodology, SOPs and testing procedures.
- c) All Existing facilities like roads, railways, water bodies, pipelines, cables and other structures likely to be affected by the construction of the flowlines shall be identified prior to commencement of construction works. Necessary safety measures to protect the pipeline and existing facilities during construction shall be implemented.
- d) Guidelines provided in OISD-GDN-192 related to safety during construction must be followed.
- e) All construction related records of pipeline shall be maintained in physical/ digital format for a minimum period of 7 years.
- f) GIS mapping of flow lines shall be done.

6.2 Survey of Pipeline route:

Pipeline route should be finalized considering following aspects:

- a) Approachability, ease of construction, operation and maintenance.
- b) Protection of environment, other properties and facilities
- c) Electrical interference.
- d) Perennially marshy, waterlogged area.
- e) Population density.
- f) Ecologically sensitive area.

Pipeline should not be located within 15.0 metres of any private dwelling or any industrial building or place of public assembly in which persons work, congregate or assemble, unless it is provided with at least 300 mm of additional cover or other mitigation measures like protection with RCC slab.

6.3 Welding:

- a) Prior to start of ~~production~~ welding, a pipe welding procedure shall be established, qualified and recorded in accordance with as per latest edition of API 1104 or ASME Section IX or equivalent standard.
- b) Welder qualification shall be performed & recorded as per latest edition of API 1104 or ASME Section IX or equivalent standard.
- c) Welder re-qualification shall be required if the welder has not performed any welding for the welding process qualified for the past 6 months. The welding procedure followed during the qualifying tests shall be recorded in detail.
- d) The weld joints shall be numbered and marked along with the welder identification, adjacent to the weld joint on the progressive direction of main line.
- e) The open ends must be capped or plugged at end of the day when welding activity is stopped, to prevent the entry of dirt, dust, water or any foreign matter into the line.

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- f) Prior to cutting, grinding and welding in areas in which the possible leakage or likely presence of vapour or flammable liquid constitutes a hazard of fire or explosion, a thorough check shall be made to determine the presence of a combustible gas mixture or flammable liquid. Cutting, grinding and welding work shall begin only when safe conditions are indicated and a valid hot work permit as per OISD-STD-105 is issued by Authorised person.

6.4 Tie- in welding:

- a) The tie-in of two sections shall be carried out only after the section has been welded and lowered and preferably back filled except for the ends which are to be joined which shall be kept on skids or kept in the ditch and shaded.
- b) The tie-in operation includes necessary cutting, bevelling, line-up, cleaning, welding, coating and wrapping, lowering in, backfilling for the tie-portion as per standard practice. Necessary widening of the trench should be carried out as per requirement.
- c) The tie-in should be done in such a way as to leave no strain in the pipe.

6.5 Welding Inspection:

- a) The quality of each weld shall be examined by visual inspection. Regardless of operating hoop stress, carbon steel butt welds in hydrocarbon pipeline shall be subjected to radiographic examination The number of welds normally to be examined shall be as follows:

Minimum 10% (ten) of total welds in case of liquid pipelines
Minimum 20% (twenty) of total welds in case of gas pipelines.

In case of specific requirement of respective process fluid/ service, extent of radiography may be higher than as mentioned above.

100% (one hundred) for all critical locations such as highways, railways, rivers, nallahs, major water bodies, tie-ins including in-plant piping etc.

- b) All weld inspection shall be done as per provisions of API 1104 or ASME Section IX or equivalent standard. Authorization for repair of welds, removal and repair of weld defects and testing of weld repairs shall be in accordance with API 1104 or ASME Section IX or equivalent standard.

6.6 Excavation:

- a) In cultivable land and other specifically designated areas, the top 300 mm soil excavated from the pipeline trench shall be stored separately. This topsoil shall be replaced in original position after backfilling and compacting of the rest of the trench.
- b) The width of trench shall be such that a minimum clear distance of 200 mm for trench in normal soil and 300mm for trench in rock is maintained between edge of pipe and the trench wall at the bottom of trench.

6.7 Lowering:

- a) Before lowering operations are commenced, pipeline section shall be visually inspected for damages, dents or other defects. Defect, if any, shall be completely repaired. Particular

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attention shall be paid to the suitability of the trench to allow the pipeline to be lowered without damaging the protective coating and to give a reasonably even support to the pipeline.

- b) Before lowering, a complete check by holiday detector set at an appropriate voltage shall be carried out. All protective coating defects and damages identified by holiday detection shall be repaired. Temporary end caps/plugs shall be provided at ends of pipe section prior to lowering of the section in the trench so as to prevent ingress of water, dirt, foreign material etc.

6.8 Backfilling:

Backfilling shall be carried out immediately after the pipeline has been laid in the trench. In cultivable land and other specifically designated areas, topsoil excavated from the trench of the Right-of-Use shall be stored separately and shall be restored to normal condition. The backfill material shall contain no extraneous material and/or hard lumps of soil, which could damage the pipe and/or coating or leave voids in the backfilled trench. Where the ditch is flooded, care shall be exercised so that the pipe is not floated from the bottom of the ditch. In rocky areas and areas with hard soils / gravels, minimum 150 mm thick padding of soft soil / sand shall be provided all around the pipe. In rocky areas, protective layer of rock-shield / rock guard shall be provided in addition to soft padding to prevent damage to coating / steel pipe during installation.

6.9 Hydro testing of Flow lines:

Flow lines (including underground and above ground portion) shall be tested to hydrostatic pressure test. The minimum test pressure at any point along the pipeline shall be as 1.25 times the internal design pressure. The test pressure shall be maintained for a minimum period of twenty-four (24) hours except where specifically mentioned. Pressure variations during testing shall be acceptable, if caused by factors other than leakage, like temperature variations. Maximum unaccounted pressure variation shall not exceed 0.3 bars. Pipeline not meeting the requirements shall be repaired if necessary and retested in accordance with the requirements of this standard after addressing the reasons for variations. Typical Format for 24 hours hold (Hydrostatic test) is provided in Annexure-II.

Respective clause on hydrotesting of mainline as per OISD-STD-141/ OISD-STD-226 may be referred for further details.

6.10 Bleeding of Hydro tested section:

Once hydrotesting is completed, depressurization and dewatering shall follow the plan included as part of the pressure test procedure. Draining of test fluid by means of pigging using inert media or any other suitable methodology, shall be ensured. Flow and pressure release should be monitored. Test medium removal lines should be properly anchored and compatible with the pressures expected during the removal of the test medium. The release point(s) should be fixed in vertical position. Typical depressurization rate should not exceed 5 psi/minute.

7.0 OPERATION, INSPECTION & MAINTENANCE OF FLOW LINES

- a) The operator shall develop standard operating procedure/ manual which shall include as a minimum following:
 - i) System Description.

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- ii) Detailed well-wise flow line layout with locations and GIS mapping of flow lines.
- iii) Operation and maintenance including shutdown/ isolation procedures.
- b) Flow lines that are in shut in conditions / not flowing fluids, a tag out device to each pipeline shall be provided.
- c) Before completion of design life of flow lines, operator shall carry out thickness measurement at minimum 2 identified locations for each kilometre length of the flow lines. For thickness measurement, guidance of OISD-STD-130 may be followed. The locations should be identified based on flow line elevation, change of soil conditions, high tension overhead power line in vicinity. Thickness survey shall be carried out at every five (5) year interval. Data of such thickness survey shall be reviewed with respect to previous/ baseline data and corrective actions shall be taken accordingly. The integrity of the pipeline shall be established after carrying out thickness survey.
- d) In case Integrity of the pipeline could not be established by thickness survey as per section (c) above, pressure testing of flow lines shall be carried out at test pressure of 1.1 times of desired maximum operating pressure (limited to design pressure). The holding period of such test shall not be less than 4 hours. Acceptability of the test shall be in accordance with clause 6.9(b) of this standard. Pressure testing shall be carried out at every 5 (five) year interval.
- e) After completion of design life of flow lines, thickness measurement/ pressure testing (as applicable) shall be carried out at three (3) years interval. In case of flow lines where risk-based inspection (RBI) system is implemented, thickness inspection survey frequency shall be as per RBI requirements.
- f) After carrying out any well intervention involving well closure production from well, production shall be resumed only after pressure test at operating pressure of all connected flow lines has been carried out.
- g) Wherever coated pipes have been laid, coating surveys shall be carried out as per OISD-STD-188. PSP monitoring shall be carried out as per OISD-STD-188, wherever CP system has been provided.
- h) The flange joint in flow lines shall be provided with either spiral wound metallic gaskets or metallic ring type gaskets, depending upon the piping class. Plain asbestos sheet / reinforced gaskets shall not be used.

8.0 FLOWLINE REPAIR:

- a) All repairs shall be undertaken in a planned manner, such plan shall among other things, include safety during repair. Maintenance / repair plan shall be approved before executing actual repair work. Only sufficiently experienced/ trained personnel are permitted to supervise and undertake repair. All repair methodology shall fall within the limits permitted by the relevant code like ASME B31.4/ PRCI Pipeline Repair Manual/ ASME PCC2/ equivalent standard widely accepted in the pipeline industry or Pipeline Repair Manual developed in line with the abovementioned guidelines and approved by the management. Precautionary measures and procedures outlined in API Publ. 2200 should be followed.

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- b) Repair welding procedures and welders performing repair work shall be qualified in accordance with API standard 1104 or section IX of the ASME Boiler and Pressure vessel code.
- c) Flow line shall be repaired by any one of the following:
 - I. Mechanical Bolt-on-Clamp: Repairs may be made to both leaking and non-leaking defects by the installation of a mechanically applied clamp. A mechanical clamp shall have a design pressure of not less than that of the pipe being repaired. Mechanical clamps shall not be used to repair circumferentially oriented defects unless designed to withstand the axial load.
 - II. By installing full encirclement welded split sleeves to contain internal pressure and shall have a design pressure of not less than the maximum allowable operating pressure. This shall be fully welded both circumferentially and longitudinally. Length of full encirclement split sleeves shall not be less than 100mm and not more than 2000 mm.
 - III. By cutting out cylindrical piece of pipe containing the defect and replacing the same with a pretested pipe of minimum 2-meter length meeting the required pipe specification.
 - IV. In case of repair of coated pipe, all damaged coating shall be removed and new coating shall be applied to replacement pieces of pipe, welded patches and full encirclement welded split sleeves used in making repairs. It is permitted to use a different variety of coating that is not similar to the coating in the existing pipe/ portion of the pipe that has been repaired or replaced. However, compatibility among the variety of coating may be ensured.
 - V. Composite sleeves/ wraps may be used to repair defects that have been removed by grinding. Composite sleeves shall not be used to repair leaks, metal loss with a depth greater than 80% of the nominal wall thickness, metal loss due to internal corrosion, cracks, or circumferentially oriented defects.
- d) The history of the leakages and repairs/ installed clamps in the flow line shall be maintained.
- e) Policy shall be developed towards timeline for replacement of pipeline/ pipe section that has developed multiple leakages and associated repairs and same shall be followed.

8.1 Testing of Repair in Flowline

- a) When a repair in a pipeline is made by cutting out a section of the pipe as a cylinder and replacing it with another section of pipe, the replacement section of pipe shall be pretested to a pressure equal to 1.25 times the design pressure of the parent pipeline for minimum period of 4 hrs. prior to installation. Radiographic inspection or other NDT technique like dye penetrant test etc, shall be carried out on all tie-in-butt weld joints after the installation.
- b) In case of pipeline leakage repair is attended by welding full encirclement sleeve or mechanical clamp, the pipeline shall be pressurised to maximum operating pressure before start of operation to check integrity of sleeve/ clamp. Holding period for this test shall be

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minimum one hour. Mechanical clamp shall be welded only after successful pressure test as indicated above.

9.0 Abandonment of Flow Lines:

Abandonment of flow lines shall be carried out as per clause no. 22 of OISD-STD-141.

10.0 Preservation of Flow Lines:

Flow lines which are in shut in conditions and are not abandoned considering use in future, pipeline shall be preserved using water dosed with adequate quantity of corrosion inhibitors or by filling the line with non-flammable non-toxic gas (nitrogen) and at a positive pressure, as applicable.

11.0 INSPECTION AND INTEGRITY ASSESSMENT OF ONSHORE NON PIGGABLE METALLIC CROSS COUNTRY PIPELINES WITH PIPELINE SIZES 4” NPS AND ABOVE.

- a) Integrity of non piggable cross-country pipeline with pipeline length less than 10 KM shall be established by methods like Direct Assessment, Large Standoff Magnetometry based techniques, long range ultrasonic thickness testing or any other suitable NDT technique. For ultrasonic thickness testing, clause No. 8.1.2 of OISD-STD-130 to be referred. In this case frequency of inspection shall be 5 years. A comparative study of the thickness measurements shall be done with the previous set of readings and appropriate measures shall be taken as per the operator's integrity management plan.
- b) In case Integrity of the pipeline could not be established by using any of the methodology as per section (a) above, pressure testing of above pipelines shall be carried out at test pressure of 1.1 times of desired maximum operating pressure (limited to design pressure). The holding period of such test shall not be less than 4 hours. Acceptability of the test shall be in accordance with clause 6.9(b) of this standard. Pressure testing shall be carried out at every five (5) year interval.

12.0 Other types of Pipes for upstream hydrocarbon industries:

Currently, mechanically lined pipes of carbon steel material with internal liner (as per API 5LD and API 5LC) and non-metallic pipelines are also being used globally, in upstream hydrocarbon onshore operations according to various API and other equivalent standards.

Hence, in addition to conventional type of steel pipes currently being used by Indian industries, any other type of mechanically lined pipes of carbon steel material with internal liner, non-metallic (spoolable/non spoolable) pipe may be used as flow lines in upstream hydrocarbon operations subject to regulatory process, as applicable. Details of one such spoolable non-metallic pipe is given in Annexure-1 for reference. If any such pipes are used by industry, its manufacturing, laying, operation maintenance, inspection and integrity of pipelines shall be as per applicable/ recognized international/ national standards

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13. Reference:

Following standards, codes and publications have either been referred to or used in the preparation of this document and this standard shall be read in conjunction with the same.

- I. OISD-STD-141: Cross Country liquid hydrocarbon pipeline
- II. OISD-STD-214: Cross Country LPG Pipelines
- III. OISD-STD-226: Natural Gas Transmission Pipelines and City Gas Distribution Networks
- IV. OISD-STD-130: Inspection of Piping Systems
- V. API Specification 15S: Spoolable Reinforced Plastic Line Pipe
- VI. API RP 15SIH: Installation and Handling of Spoolable Reinforced Line Pipe
- VII. API RP 15SA: Integrity Management of Spoolable Reinforced Line Pipe
- VIII. API 17B: Recommended Practice for Flexible Pipe
- IX. API 17J: Specification for Unbonded Flexible Pipe
- X. API RP 1110: Pressure Testing of Steel Pipelines for the Transportation of Gas, Petroleum Gas, Hazardous Liquids, Highly Volatile Liquids or Carbon Dioxide
- XI. ASME B 31.4: Pipeline Transportation System for Liquids and Slurries
- XII. ASME B 31.8: Managing System Integrity of Gas Pipelines
- XIII. API-RP-1104: Welding of Pipelines and Related Facilities

Annexure-I

Onshore Non-Metallic Pipelines for Upstream Hydrocarbon Industry

1.0 INTRODUCTION

This annexure provides requirements for the design, construction, operation, inspection and maintenance of spoolable reinforced plastic line for use as flow lines in upstream hydrocarbon operations under sour/non-sour service applications with pipeline sizes 2" NPS and above, laid between wellhead and installation for transportation of well fluid, water/effluent disposal, gas injection, lift gas, water and polymer injection. These pipelines shall be in accordance with API 15S, API RP 15SIH, API RP15SA, API 17 B, API 17J and (is required in spoolable) unless specified by owner.

2.0 DEFINITIONS

2.1 Blistering:

Damage in the form of gas-filled pockets caused by the release of absorbed gas on depressurization within a solid polymer layer (e.g., polymeric liner).

2.2 Brittle Failure:

A failure mode that exhibits no visible (to the naked eye) permanent material deformation (stretching, elongation, or necking down) in the area of the break.

2.3 Connector:

A device used to provide a leak-tight (except to intentionally vent gas) structural connection between the end fitting and adjacent piping (e.g., bolted flanges, clamped hubs, and proprietary connectors).

2.4 Coupling:

A fitting developed for joining one section of pipe to another. (e.g., in-line connector)

2.5 Cover:

The protective outer sheath of the pipe.

2.6 Cyclic Loading:

More than 7000 cycles and $\Delta P/NPR > 6\%$, where ΔP is maximum to minimum amplitude.

2.7 Disbondment:

Separation of the previously bonded layers.

2.8 End Fitting:

A mechanical device that forms the transition from the pipe to the connector.

2.9 Field Fitting:

An end fitting or coupling designed for permanent installation.

2.10 Liner:

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A continuous polymeric layer that is in contact with the conveyed fluid.

2.11 Purging:

It is the process of replacing the atmospheric air within a container (pipeline, vessels, filters etc) by an inert substance in such a manner so as to prevent the formation of explosive mixture.

2.12 Spoolable Pipe:

Pipe that can be provided as a coil or on a structural reel for transportation. For the purposes of this specification, the terms coils, reels, and spools may be used interchangeably.

2.13 Spoolable Composite Pipe:

A family of composite reinforced pipes in which the flexible structural layer enables spooling and unspooling. This includes, but is not limited to, S-GRE and RTP pipe.

2.14 Spoolable Glass Reinforced Epoxy Pipe (S-GRE):

A type of spoolable composite pipe in which the structural layer consists of an even number of balanced helical windings of continuous glass fibers in an epoxy thermoset resin matrix.

2.15 Spoolable Reinforced Thermoplastic Pipe (RTP):

A type of spoolable composite pipe that consists of a thermoplastic liner on which the structural layer is wound, consisting of an even number of balanced helical windings of reinforcement elements.

2.16 Tape:

A reinforcement type in which the strength elements are consolidated in a matrix material and provided as a flat, flexible, ribbon.

2.17 Traceability:

The ability to identify the origin of materials and parts used to manufacture a product and/or the product processing or manufacturing history.

3.0 DESIGN**3.1 GENERAL**

3.1.1 Such pipes shall be manufactured in accordance with API 15S or other equivalent standard. The manufacturer(s) shall have valid license from API authorities to manufacture pipes as per API 15S.

3.1.2 This specification is confined to pipes, end-fittings, couplings and does not relate to other system components and appurtenances. Where other system components (e.g. elbows, tees, valves) are of conventional construction they will be governed by other applicable codes and practices.

3.1.3 The pipelines shall be designed in a manner that ensures adequate public safety under all conditions likely to be encountered during installation, testing, commissioning and operating conditions. All materials and equipment shall be selected to ensure safety and suitability for the condition of use.

3.1.4 A design Life of minimum 25 years for flow lines system should be considered by the owner for designing various system and facilities. The life of pipeline can be extended beyond the design life subject to satisfying the comprehensive pipeline integrity assessment as specified by the manufacturer.

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- 3.1.5** While designing the pipeline system, reasonable protection shall be provided to prevent damage to the pipeline from unusual external conditions.

3.2 MATERIAL SELECTION

- 3.2.2** Spoolable reinforced plastic pipe consist of a liner with helically wrapped steel or non-metallic reinforcing elements and another cover. The helical reinforcing elements shall be of a single material. Additional non-helical reinforcing elements are acceptable. The spoolable reinforced line pipe under this specification shall be capable of being spooled for storage, transport and installation. Spoolable pipeline comprises of three layers as below:

3.2.3 Liner (innermost layer):

Liner is the continuous polymeric layer that is in contact with the conveyed fluid. The liner shall be confirmed through testing to retain its integrity for the specified fluids under the given service conditions. The final material properties after processing shall be within the acceptable range of properties for the given service conditions and verified based on the material qualification program.

3.2.4 Reinforcement layer:

It is the intermediate layer of the pipe. It comprises of matrix material and reinforcing element. Reinforcing elements is the primary contributor to the hydrostatic strength of the pipe and it may be metallic or non-metallic.

3.2.5 Cover (outermost layer):

Cover is the protective outer sheath of the pipe. The cover material shall sustain its function for the service conditions.

3.3 DIMENSIONS

The nominal diameter for sizes 3 inches and greater shall be in half-inch increments and the minimum allowable inside diameter shall be equal to or greater than the nominal diameter minus $\frac{1}{2}$ inch.

For nominal diameter smaller than 3 inches, the minimum allowable inside diameter shall be equal to or greater than the nominal diameter minus $\frac{1}{4}$ inch.

The manufacturer shall document both the internal and external diameter of the pipe. Sizing in Accordance with Nominal Diameter shall be as below:

Nominal Diameter (in.)	Minimum ID (in.)
2	1.75
2.5	2.25
3	2.5
3.5	3.0
4	3.5
4.5	4.0
5	4.5
6	5.5

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8	7.5
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The tolerances for the outside diameter of the pipe and the thickness of each layer of the pipe shall be verified during design stage.

Qualification program, Process and Quality assurance requirements and Marking and Documentation shall be in accordance with API 15S.

4.0 HANDLING & INSTALLATION OF SPOOLABLE REINFORCED LINE PIPE

- 4.1** The spoolable reinforced line pipe shall be handled & installed as per API 15SIH.
- 4.2** If termination of pipe has to be done in an area which is accessible to general public or has a possibility of third-party damage, reinforced pipe shall be terminated below ground and connected to the surface facilities/ wellhead through metallic riser. Reinforced pipe shall be terminated at a minimum safe distance from field manifold, while a minimum distance of 30 meters shall be maintained from well head. Surface markers shall be provided at termination point for protection of non-metallic pipe to ensure that it is not damaged during any maintenance activities.
- 4.3** Metallic tracer wire and warning tape shall be laid above the buried pipeline to locate pipe. Wherever caution signs are installed, these shall be in bilingual (local language and Hindi / English).
- 4.4** The location of a new buried pipeline, when running parallel to an existing buried pipeline, shall be at a minimum clear distance of 5.0 meters from the existing buried pipeline when heavy conventional construction equipment is expected to be utilized. This distance may be reduced after careful assessment of construction methodologies so that it does not result in unsafe conditions during construction. In any case the minimum clear distance shall not be less than 3.0 meters. These areas shall be distinctly identified on the ground during construction.
- 4.5** When laying more than one new pipeline in the same trench, a minimum clear horizontal separation distance of 500 mm shall be maintained.
- 4.6** When pipeline has to cross any existing underground pipeline, cable, drain or other services, the pipeline shall be laid at least 500 mm below or above such existing utility / services in a manner that will not obstruct access to such services for inspection, repair, or maintenance.
- 4.7** Pipeline should not be located within 15.0 metres of any private dwelling or any industrial building or place of public assembly in which persons work, congregate or assemble, unless it is provided with at least 300 mm of additional cover or other mitigation measures like protection with RCC slab.
- 4.8** In case the pipeline is to be laid parallel to roads/highways, effort shall be made to locate the pipeline outside the limits of highway right of way.
- 4.9 Pipeline Cover**

All hydrocarbon pipelines shall be buried below the ground level in line with the “Minimum Cover for Buried Pipelines”, as specified below:

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Minimum Cover Requirements for Non-metallic Pipelines

Sl. No.	Location	Minimum Cover (m)
1.0	Industrial, commercial, residential, cultivated, agricultural and barren areas	2.0 (For normal soil) 1.5 (For rocky strata)
2.0	Minor water crossing/ canal/ drain/ nala	2.0 (For normal soil) 1.5 (For rocky strata)
3.0	HDD crossing rivers (below scour depth)	2.5
4.0	HDD crossing at lined Canals (below bed)	4.0
5.0	River crossings (below scour) –open cut	2.5 (For normal soil) 1.5 (For rocky strata)
6.0	Cased/uncased road crossing/ Station approach	2.0
7.0	Railway crossing	1.7
8.0	Marshy land/ Creek area	2.0

NOTES:

- The depth of cover shall be measured from the top of the pipe to the top of the undisturbed surface of soil or the top of graded strip, whichever is lower.
- The cover shall be measured from the top of road or top of rail, as the case may be.
- For river/watercourses that are prone to scour and / or erosion, the specified cover shall be measured from the expected lowest bed profile after scouring / erosion. When scour level is not known, an additional cover of at least 1 m (over and above the cover mentioned in Table) shall be provided from the existing bed of the river/water course except in case of rocky river bed;
- Whenever the above provisions of cover cannot be provided due to site constraints, additional protection in form of casings, bridging, etc. shall be provided.
- When insisted by authorities, the depth shall be maintained as per the direction of concerned authorities.

- 4.10** In rocky areas and areas with hard soils / gravels, minimum 150 mm thick padding of soft soil / sand shall be provided all around the pipe. A protective layer of rock-shield / rock guard may be provided in addition to soft padding to prevent damage to pipe during installation.
- 4.11** No dwelling or permanent structure in any form shall be permitted within the Right of Way (RoW) or Right of User (RoU).
- 4.12** All Pipelines shall be pressure tested in-place after construction. Hydrotesting shall be done at a

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minimum of 1.25 times the system design pressure. The test pressure shall be maintained for a minimum period of twenty-four (24) hours. API RP 15SA should be used for guidance for the hydrostatic test.

5.0 INTEGRITY MANAGEMENT OF SPOOLABLE REINFORCED LINE PIPE

5.1 Integrity management shall include threat identification, identification of potential failure modes, risk assessment, testing requirements, initial and ongoing inspection practices, end-fitting and coupling inspection, potential mitigations, repair, and associated documentation. Same shall be as per API 15SA (First edition).

5.2 Test Spool

Test spools are full-scale sections of pipe removed for testing. These test spools can either be designed into the field originally and operated in parallel, or sections of the original pipeline can be cut out and replaced. Test spool locations should be selected to reflect the severity of service conditions in terms of loading, deformation, and internal or external environmental conditions.

Periodic removal and testing of test spools from areas identified in failure mode and risk analysis of a pipeline, provide a means to evaluate the long-term performance of the spoolable pipe when subjected to the actual pipeline operating conditions.

The first testing of test spool shall be carried out after five (5) years, from commissioning but not later than Ten (10) years. However, if pipeline fails due to unattributable reasons, testing of spool from failed section shall be done immediately. Subsequent spool testing shall be carried out at every five (5) year intervals from last testing of test spool of the pipeline and test spools shall not be from same section/ location.

Acceptance criteria for the periodic test shall be discussed and agreed upon between the operator and manufacturer based on the manufacturers test data.

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MAJOR CHANGES IN REVISED EDITION

Clause No. 2:

Scope of the standard has been revised to include Design, construction, operation, inspection and maintenance of onshore non piggable metallic pipelines of upstream hydrocarbon industry (of sizes 2" NPS and above), laid between Wellhead and installation for transportation of well fluid, water/effluent disposal, gas injection, lift gas, water and polymer injection.

Scope includes Inspection and Integrity assessment of onshore metallic cross-country pipelines designed as per OISD-STD-141, OISD-STD-214 & OISD-STD-216 which are either non piggable or have become unpiggable.

Clause No. 4.0

For all new carbon steel pipe nominal wall thickness including corrosion allowance shall not be less than 7 mm.

Clause No. 5.0

New pipeline shall be externally coated with suitable anti corrosion coating e.g. 3-layer Polyethylene / Polypropylene or Fusion Bonded Epoxy coating or any other equivalent type of coating.

Clause No. 6.9

Hydrotesting requirements for new flow lines added. Typical Format for 24 hours hold (Hydrostatic test) provided in Annexure-II

Clause No. 6.10

Depressurization and dewatering requirements post hydrotesting added.

Clause No. 7.0:

Thickness measurement at minimum 2 identified locations for each kilometer length of the flow lines at every five (5) year interval added. The integrity of the pipeline shall be established after carrying out thickness survey.

In case Integrity of the pipeline could not be established by thickness survey, pressure testing of flow lines shall be carried out at test pressure of 1.1 times of desired maximum operating pressure (limited to design pressure).

After completion of design life of flow lines, thickness measurement/ pressure testing (as applicable) shall be carried out at three (3) years interval.

Clause No. 8.0:

Guidelines for repair and testing of repair in flowlines added.

Clause No. 11:

Methods like Direct Assessment, Large Standoff Magnetometry based techniques, long range ultrasonic thickness testing or any other suitable NDT technique added.

In case Integrity of the pipeline could not be established by using any of the methodology as above, pressure testing of above pipelines shall be carried out at test pressure of 1.1 times of desired maximum

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operating pressure (limited to design pressure).

Clause No. 12:

In addition to conventional type of steel pipes currently being used by Indian industries, any other type of mechanically lined pipes of carbon steel material with internal liner, non-metallic (spoolable/non spoolable) pipe may be used as flow lines in upstream hydrocarbon operations subject to regulatory process, as applicable. Details of one such spoolable non- metallic pipe is given in Annexure-1 for reference.

FUNCTIONAL COMMITTEE MEMBERS ON

GUIDELINES ON INSPECTION OF ONSHORE NON-PIGGABLE PIPELINE (2013)

S.No.	Name	Designation & Organization	Position in the Committee
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FORMAT FOR SENDING COMMENTS ON THE DOCUMENT

Comments on each clause/ sub-clause/ table/ figure etc. to be stated in a fresh row. Information/ comments should include reasons for comments, technical references and suggestions for modified wordings of the clause.

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Name of the Person with Organization: _____

Clause/ Para/ Table/ Figure no. commented	Original Wordings	Comments/ Modified Wordings	Justification for Proposed Change

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